Name of the Program: B.Sc. in CSE (94) **09 September 2021**
Semester: Winter 2020-2021 **Time: 2:30 PM – 4:00 PM**

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER Final EXAMINATION **Winter Semester: 2020-2021**
Course Code: CSE 4303 **Full Marks: 75**
Course Title: Data Structures **Time: 1 Hour 30 Minutes**

There are **03 (three)** questions. Answer all of them. Figures in the right margin indicate marks. The examination is an **Online** and **Closed Book.** Marks of each question and corresponding **CO** and **PO** are in the brackets. Renaming format 'fullID_cse4303.pdf'.

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1. Showing detailed steps, perform the following operations for an AVL tree. 10+5+5+10
$(ID + 50), 100, 175, 30, floor(\sqrt{ID}), 50, (ID\%10), 15, (ID^2), birthdate$ CO1
PO1
   i. Insert the above-mentioned numbers in the tree. [Mention the $balance_factor$ with each node in brackets]
   ii. $Successor(ID)$ [simulate properly with justification on how to reach the value]
   iii. $Delete(root)$, Replace with $Tree_maximum$.
   iv. $Inorder(root)$: Traverse the whole tree. [Simulate by showing the different stages of the recursion stack]

   Note:
   - $'ID'$ means the last two digits of your $student_id$
   - '$birthdate$' is the 'dd' from your date of birth considering $'dd/mm/yy'$ format.

2. Show detailed steps perform the following operations: 20
$(ID + 50), 100, 175, 30, floor(\sqrt{ID}), 50, (ID\%10), 15, (ID^2), birthdate$ CO1, CO2
PO1,PO2
   i. Build a $Segment\ Tree$ using the above-mentioned numbers where internal nodes will represent the Sum of values within a given range.
   ii. Increment all the values within range [2,5] by $'ID'$.
   iii. What is the summation of the range [4,9]?
   iv. Increment all the values within range [6,8] by $'floor(\sqrt{ID})'$.
   v. What is the current value of $index - 5$?
   vi.

   Note: Array index starts from 1.

3. a) i. Suppose instead of quadratic probing, we use 'cubic probing'; here the ith probe is at $hash(x) + i^3$. Does it improve on quadratic probing? 5+5
CO1, CO4
PO1,PO3
   ii. Explain the concept of '$Rolling\ Hash\ Function$' in the context of the $Rabin\ Karp\ Algorithm$. Simulate its benefit by searching the string '$abcd$' in your $full_name$. (Show only the first three attempts.)

   b) What are the benefits of using $Trie$ over $Hash\ Tables$ for storing string? 5
CO1, CO2
PO1,PO2

   c) What is Data Structure? Why do we care about it? Write an appropriate use case for each of the data structures taught in this course. 10
CO1, CO3
PO1,PO3